

Nathan Ngau

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Computer Science student with hands-on experience in multithreaded systems, file systems, and data pipelines using C and Python, seeking a software engineering internship or entry-level SWE role.

EDUCATION

B.S. Computer Science

Temple University, Philadelphia, PA

SKILLS

Programming: Python, Java, C, SQL, Bash/Shell, TypeScript

Data & ML: Pandas, NumPy, PyTorch, Regression, Feature Engineering, Data Analysis, Data Visualization, OpenCV

Systems: Linux, Multithreading, POSIX Threads, File Systems, Concurrent Programming, UML

Tools: Git/GitHub, Jupyter Notebook, Docker, Kubernetes, Figma

PROJECTS

Peso (Computer Vision Lifting Analysis App)

Tech Stack: React Native, TypeScript, Python, FastAPI, MediaPipe, OpenCV

- Built a full-stack lifting analysis app across 8 authenticated FastAPI endpoints and 26 React Native/Expo modules by implementing Supabase auth, storage, upload, queued analysis, review, export, and discard workflows.
- Generated 10+ squat metrics per video by sampling clips at up to 18 FPS with MediaPipe/RTMPose, rep segmentation, depth scoring, torso-angle tracking, velocity estimates, and coaching feedback.
- Improved core analysis reliability, validated by 64 passing backend tests, by adding pose-quality checks, landmark validation, fallback model selection, idempotent queueing, and analysis versioning.
- Designed video persistence for queued, completed, saved, discarded, expired, and exported videos by integrating Supabase rows, storage objects, signed URLs, JWT ownership checks, cleanup logic, and H.264 exports.

Restaurant Management Simulation (Multithreaded C Project)

Tech Stack: C, POSIX Threads, Mutexes, Condition Variables, Semaphores

- Built a multithreaded restaurant simulation modeling 100+ concurrent customers, 10 waiters, and kitchen workflows.
- Implemented producer-consumer queues and synchronization primitives to handle 500+ concurrent tasks efficiently.
- Prevented race conditions and deadlocks through thread-safe resource management, achieving 0 runtime crashes.

Lifting Stats (Strength Training Analytics & Forecasting)

Tech Stack: Python, Pandas, Data Analysis, Git

- Developed a data pipeline transforming 3+ years of workout logs and 10,000+ data points into performance analytics.
- Created 5+ custom metrics including estimated 1RM, weekly maxes, and training volume trends.
- Applied regression models and visualizations to forecast performance progression with an 85% predictive accuracy.

Virtual File System (FAT-Based Implementation)

Tech Stack: C, Linux, Virtual Disk I/O

- Designed and implemented a FAT-style file system supporting 4 core operations: read, write, seek, and delete.
- Built persistent storage structures for file allocation and metadata management, optimizing disk usage by 15%.
- Supported 64 files and 32 concurrent file descriptors with efficient block management.

EXPERIENCE

Building Manager | Howard Gittis Student Center

Philadelphia, PA | Oct 2025 - Present

- Managed event operations across 15 rooms accommodating up to 999 attendees, facilitating 100+ successful events per semester.
- Diagnosed and resolved technical issues during 50+ live events, reducing average equipment downtime by 30%.
- Maintained operational logs for \$50,000+ in A/V equipment, achieving 100% accuracy in maintenance tracking.

Co-owner / Manager | Manor Laundromat LLC

Lancaster, PA | Sep 2022 - Present

- Troubleshoot and repaired 40+ commercial machines to reduce downtime by 25% and improve overall operational efficiency.
- Managed payment systems processing \$10,000+ in monthly revenue and handled support for 200+ monthly patrons.
- Serviced 5 PA skill machines with weekly payout handling, maintaining 0 financial discrepancies over 2 years.

Warehouse and Retail Worker (Seasonal) | Soccer Post

Lancaster, PA | Jun 2024 - Aug 2024

- Assisted 40+ customers daily with product recommendations, contributing to a 15% increase in seasonal sales.
- Organized \$25,000+ in retail inventory and maintained merchandising displays, reducing stock misplacements by 20%.

ACTIVITIES & CERTIFICATIONS

Six Sigma White Belt Certification (2026)

AI Fluency: Framework & Foundations, Anthropic (2026)

Temple University Development Club | Association for Computing Machinery | Marketing Committee | Club Lacrosse